

Demonstration of Proposed Features

Date	12 August, 2026
Team ID	Individual
Project Name	AI Knowledge Retention Generator
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the individual ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1.	Forgetting Curve Calculator	Calculates real-time memory retention probabilities using mathematical decay models based on initial mastery and elapsed time.	Implemented	Yes	Core mathematical logic implemented in <code>utils.py</code> and validated via terminal test scripts.
2.	SQLite Storage Engine	Persists user study topics, timestamped review logs, and quiz results locally across application sessions.	Implemented	Yes	Database schema defined with complete CRUD functions for review history tracking.
3.	Gemini Dynamic Quiz Generator	Integrates Google Gemini API to generate customized multiple-choice	Implemented	Yes	Prompt templates and JSON parsing utility clean raw AI output into structured quizzes.

		questions dynamically based on decay status.			
4.	Interactive Web Dashboard	Streamlit UI providing user inputs, retention metric cards, and dynamic visual decay graphs.	Implemented	Yes	Built in <code>app.py</code> with responsive chart rendering and input forms.
5.	Automated Quiz Evaluation & Updates	Evaluates user quiz submissions in real time and automatically updates database scores and decay visualizations.	Implemented	Yes	End-to-end integration synchronizes quiz attempts directly with SQLite records.
6.	Cloud Deployment & Secrets Setup	Production deployment on Streamlit Cloud with GitHub version control and secure API key management.	Implemented	Yes	Publicly accessible live deployment configured with Streamlit Secrets Manager.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%